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Robust steering-assist torque control of electric-power-assisted-steering systems for target steering wheel torque tracking☆

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ABSTRACT

This paper describes robust steering assist torque control of electric-power-assisted-steering (EPAS) systems for a target steering wheel torque tracking. The steering assist torque control algorithm has been developed to overcome the major disadvantage of the conventional method of time-consuming tuning to achieve the desired steering feel. A reference steering wheel torque map was designed by post-processing data obtained from target performance vehicle tests with a highly-rated steering feel for both sinusoidal and transition steering inputs. Adaptive sliding-mode control was adopted to ensure robustness against uncertainty in the steering system, and the equivalent moment of inertia damping coefficient and effective compliance were adapted to improve tracking performance. Effective compliance played a role in compensating the error between the nominal rack force and the actual rack force. The performance of the proposed controller was evaluated by conducting computer simulations and a hardware-in-the-loop simulation (HILS) under various steering conditions. Optimal steering wheel torque tracking performances were successfully achieved by the proposed EPAS control algorithm.

1. Introduction

Over the past decade, steering torque feedback and steering feel performance have been vigorously researched to improve on-center handling, and quantification of vehicle steering feel and responsiveness. Bertollini et al. [1] showed that drivers' preferences f0or steering wheel feel change with vehicle speed, and Green demonstrated that a steering effort that changes with vehicle speed is preferred by drivers [2]. Several techniques have been proposed to determine steering feel according to the hysteresis curve [3]. In addition, numerous studies have attempted to combine subjective and objective criteria. The correlation of subjective and objective methods has been investigated to identify physical parameters that are associated with each other [4]. Recently, Jang et al. [5] presented a correlation of subjective and objective measures of on-center handling performance of a vehicle by utilizing an interaction formula obtained from a statistical model that relates driver ratings and key physical parameters. This paper presented the three most influential, large parameters with a statistical correlation through a steering torque and lateral acceleration curve. In this case, the absolute value of the steering torque is defined by the 'looseness' at 0 g of lateral acceleration. In addition, the torque gradient at that part is the 'steering stiffness', and the instantaneous minimum rate in the range of 0.1 g of lateral acceleration is defined by the 'sharpness on center.' More recently, a study was also performed about the factors that strongly affect steering torque feedback [6]. Despite the substantial amount of extant research on measuring factors that affect steering feel, time-consuming tuning tasks are still required to be applied to real vehicles to utilize these findings. Therefore, developing a common electric-power-assisted-steering (EPAS) system presents challenges in implementing the steering feel that the developer desires.

Oh et al. [7] described the design of a controller for a steer-by-wire system using a reference torque map. The reference torque map utilized the steering wheel angle and vehicle speed information. Kim et al. [8] presented a control logic with PID control, which comprises a return and active damping function to reduce steering torque effort for an EPAS system. In this research, improvement of return-to-center performance was proposed employing a reference steering map defined from vehicle speed and steering wheel angle. Oh et al. and Kim et al. [7,8] proposed a reference torque map to implement or improve steering wheel torque. However, it is difficult to consider the damping or friction component because the steering angular velocity is not considered in the map. Actually, the steering wheel torque feel

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possesses a hysteresis affected by friction and damping components. Kim et al. [8] additionally considered active damping control to avoid oscillation of steering wheel angle. However, the active damping control can act as a hindrance in implementing the desired steering wheel torque.

In the early 1950s, sliding mode control (SMC) was first proposed, and it was successfully implemented to tackle system uncertainties and external disturbances with good robustness [9–11]. In general, an SMC requires an appropriate control law, such that the sliding mode is reached in a finite amount of time. However, no simple SMC, regardless of whether such uncertainty or disturbance is bounded, nor adaptive sliding mode control (ASMC), has been proposed [12,13]. Moreover, integral augmented sliding mode control has been introduced to improve the control performance of a system [14,15].

This paper proposes a novel EPAS assist motor torque control to overcome the conventional time-consuming tuning method for EPAS development by introducing a reference steering wheel torque surface defined by steering angle, angular velocity, and vehicle speed. The proposed algorithm offers expandability to implement various steering wheel torque conditions and different vehicle speeds according to the user's preferences through modifying the reference map. An adaptive sliding mode control algorithm was also proposed for robustness against uncertainties of the steering system parameter to adapt external tire forces. The performance of the proposed algorithm has been investigated via computer simulation and hardware-in-the-loop-simulation (HILS) to efficiently evaluate real-time performance.

2. Dynamic model of electric power assisted steering system

A column type electric-power-assisted-steering (CEPAS) model is utilized to design the steering assist torque control law. CEPAS is one of the steering system types that is assisted by electric motor power on the column. Zhang et al. presented a mathematical model and characteristic curves of an EPAS system [19]. The steering system comprises a steering wheel, upper column, lower column, electric motor, and steering rack, as shown in Fig. 1.

Yih et al. [17] represented the dynamics of the steering system utilizing a simple second-order model to modify the vehicle handling characteristics via steer-by-wire. Yih proposed that the frequency response of the system can be approximated by using the empirical transfer function estimate (ETFE), i.e., the ratio between the output and input discrete Fourier transform (DFT). Yih assumed that the transfer function could represent the steering system, including the total moment of inertia, the effective viscous damping coefficient, the Coulomb

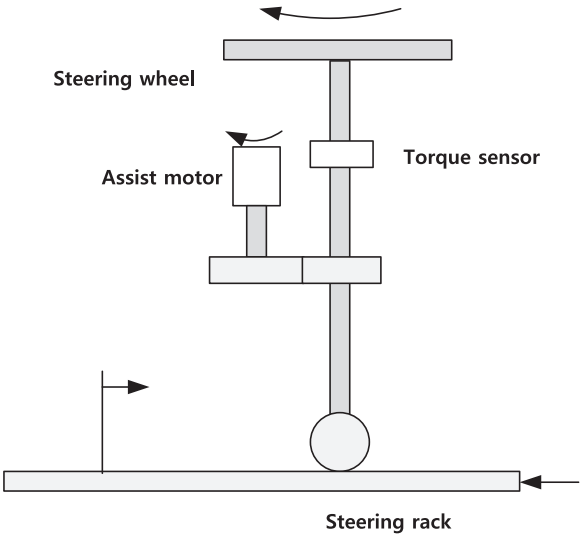


Fig. 1. Schematic of a column-type electric power assisted steering system.

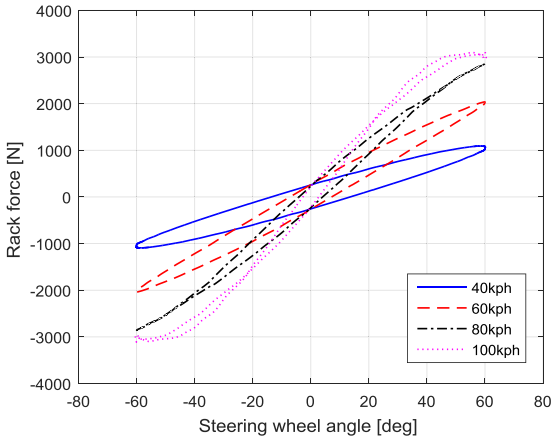


Fig. 2. Comparison of different vehicle speeds and rack force.

friction, and the disturbance from the tire force and aligning moment.

Therefore, the simple second-order model is proposed as follows:

$$J_{eq}\ddot{\theta}_{sw} + B_{eq}\dot{\theta}_{sw} = T_{sw} + T_m - K_{eff}\theta_{sw} - R_{rp}F_r \quad (1)$$

where J_{eq} and B_{eq} are the system equivalent moment of inertia and damping coefficient, respectively; T_{sw}, T_m, K_{eff}, and F_r are steering torque, assist torque, effective compliance, and rack force, respectively; and R_{rp} is the gear ratio between the rack and the end of the steering column. Most of the disturbance in steering systems is the rack force, F_r, from tire forces. Hence, rack force information is critical to implement the controller for steering systems. Gillespie [16] reported that the steering system could be represented by effective compliance in the low-frequency range. The following experiments were conducted to identify the characteristics of rack force. Fig. 2 shows the rack forces under different vehicle speeds on 0.85 road friction with weave tests at 0.2 Hz and 60° open-loop command using Carsim software E-Class Sedan. As can be seen in Fig. 2, the rack force increases linearly as the steering wheel angle increases. In addition, the rate of the rack force corresponding to the steering wheel angle increases as the vehicle speed increases. As a result, the rack force has linearity until the tire forces are saturated. Therefore, the rack force was assumed to be linear using effective compliance, K_{eff}, and selected as an adaption parameter. Effective compliance plays a role in compensating for the error between the nominal rack force and the actual rack force.

3. Target steering wheel torque tracking control

The electric-power-assisted-steering systems (EPAS) algorithm function consists of three parts, as shown in Fig. 3. The assist torque control uses the sensor information of EPAS and vehicle chassis. The target steering wheel torque map is designed by the steering angle, steering angular velocity, and longitudinal velocity of the vehicle. The nominal rack force is calculated using the linear tire model with the vehicle sensor information, and steering states with a Kalman filter are estimated to implement the adaptive sliding mode control (ASMC). Finally, the ASMC controller is used for the target steering wheel torque tracking. The ASMC calculates the assist torque input of the EPAS of the steering system at every time-step.

3.1. Target steering torque map generation

The steering characteristics are measured by conducting various tests. Sinusoidal and transition tests constitute typical test scenarios for steering characteristics. The sinusoidal test, which is also known as a weave test, measures the characteristics of the steering system, such as the degree of hysteresis and the assist torque. The transition test is carried out under constant steering angular velocity to measure the

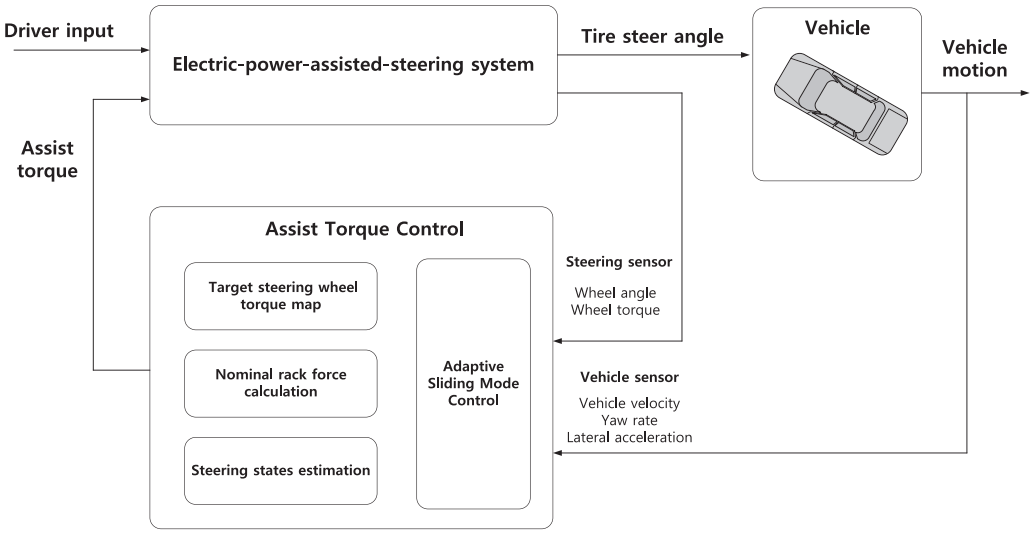


Fig. 3. EPAS control algorithm for the target steering wheel torque tracking.

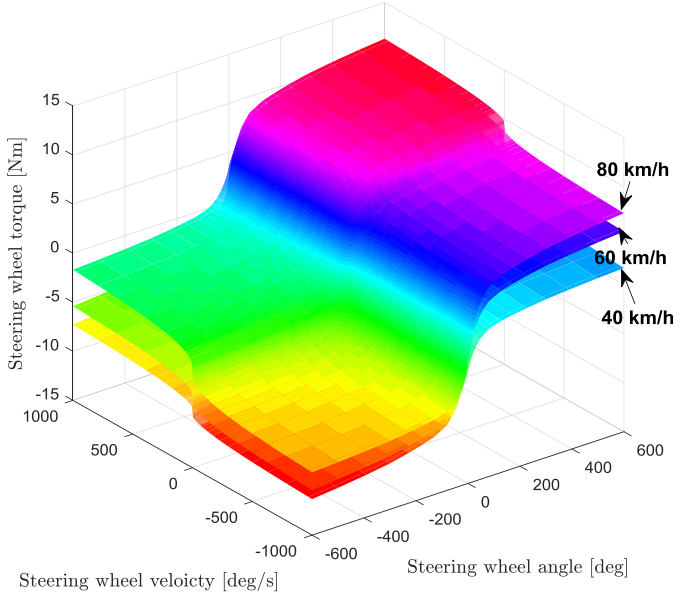


Fig. 4. Target steering wheel torque surface.

initial assist torque through the EPAS logic or the characteristics of the vehicle dynamics. These characteristics of tests are utilized to determine the target steering torque map from the steering wheel angle, steering angular velocity and vehicle speed, as shown in Fig. 4. Therefore, the damping or friction component should be considered when designing a target steering wheel torque. This component also implies that the target steering wheel torque exists, regardless of the steering angle, as long as the steering wheel has velocity. Consequently, the target steering wheel torque is created as a friction model to make a hysteresis to enhance on-center handling.

The target steering wheel torque is presented at 40, 60, and 80 km/h. If a developer desires to alter the speed of the vehicle, multiplication on low-gain at low vehicle speed and high-gain at high vehicle speed could be applied, as Oh et al. [7] proposed. Hence, utilizing the target map will not only achieve the desired steering wheel torque, but will also enhance steering feel, which is highly beneficial for the driver.

3.2. Adaptive sliding mode control design for target steering wheel torque tracking

As mentioned above, the steering dynamics in Eq. (1) are used for the control law. However, this model inevitably introduces some approximations and uncertainties, as shown in Eq. (2):

$$(\hat{J}_{eq} + \Delta J_{eq})\ddot{\theta}_{sw} + (\hat{B}_{eq} + \Delta B_{eq})\dot{\theta}_{sw} = T_{sw} + T_m - (\hat{K}_{eff} + \Delta K_{eff})\theta_{sw} - R_{rp}\hat{F}_r - \Delta_{un} \quad (2)$$

where $\hat{\bullet}$, and$\Delta\bullet$ are the nominal value and uncertainty of the value, respectively; and Δ_{un} is an unmodeled term of the steering system. To account for uncertainties, the model uncertainties and nonlinearities are assumed to be as shown in Eq. (2). This assumption is referred to as the matching condition [11]:

$$\hat{J}_{eq}\ddot{\theta}_{sw} + \hat{B}_{eq}\dot{\theta}_{sw} = T_{sw} + T_m - \hat{K}_{eff}\theta_{sw} - R_{rp}\hat{F}_r + L(t) \quad (3)$$

where $L(t)$ denotes the lumped uncertainty that is bounded, but unknown; $|L(t)| \le L_m$; and L_m are positive real constants. $L(t)$ is defined by:

$$L(t) = -\Delta J_{eq}\ddot{\theta}_{sw}(t) - \Delta B_{eq}\dot{\theta}_{sw}(t) - \Delta K_{eff}\theta_{sw}(t) - \Delta_{un} \quad (4)$$

The sliding surface can be improved by introducing an integral action into the sliding surface for steady-state accuracy, defined as [18]:

$$e(t) = T_{sw}(t) - T_{sw.des}(t) \quad (5)$$

$$s(t) = \lambda_1 e(t) + \lambda_2 \int_0^t e(\tau)d\tau \quad (6)$$

where λ_1 and λ_2 are positive gains; and $T_{sw.des}$ is the steering feel torque from the four-dimension map. The appropriate gains are selected to enhance performance. The sliding controller aims to design a control law that efficiently accounts for both parameter uncertainty and the presence of unmodeled dynamics [9]. The sliding control has continuous and discontinuous parts, in which the continuous part is the equivalent control assuming the lumped vector L in (3), and the discontinuous section compensates for uncertainties and nonlinearities.

The derivative of sliding variable $\dot{s}$ given by:

$$\dot{s}(t) = \lambda_1 \dot{e}(t) + \lambda_2 e(t) \quad (7)$$

neglecting parametric uncertainties and friction, and utilizing the nominal parameters in Eq. (3). As a result, the equivalent control u_{eq} is obtained as follows:

$$u_{eq} = \hat{J}_{eq}\ddot{\theta}_{sw} + \hat{B}_{eq}\dot{\theta}_{sw} + \hat{K}_{eff}\theta_{sw} + R_{rp}\hat{F}_r - T_{sw.des} + \frac{\lambda_1}{\lambda_2}\dot{e} \quad (8)$$

In order to guarantee the robustness of the controller against parametric uncertainties, a switching control input u_N is introduced as:

$$u_N = k \text{ sat } [s(t)/\Phi] \quad (9)$$

where k refers to the upper bound of the uncertainty, i.e., $|L_m| \le k$; and Φ is a positive constant that defines the thickness of the boundary layer that affects steady-state accuracy and robustness. Under this condition, the system can be expected to be stable, and $sat[s(t)/\Phi]$ denotes the saturation function, defined as:

$$\text{sat}[s(t)/\Phi] = \begin{cases} 1 & if\ s(t) > \Phi \\ -1 & if\ s(t) < -\Phi \\ s(t)/\Phi & else \end{cases} \tag{10}$$

Lastly, where $\hat{J}_{eq}$, $\hat{B}_{eq}$, and $\hat{K}_{eff}$ denote the adjusted parameter for compensating uncertainties in (8), whose adaptation laws are given by:

$$\dot{\hat{J}}_{eq}(t) = \frac{1}{\rho_J} s(t) \cdot \ddot{\theta}_{sw}(t) \tag{11}$$

$$\dot{\hat{B}}_{eq}(t) = \frac{1}{\rho_B} s(t) \cdot \dot{\theta}_{sw}(t) \tag{12}$$

$$\dot{\hat{K}}_{eff}(t) = \frac{1}{\rho_K} s(t) \cdot \theta_{sw}(t) \tag{13}$$

where ρ_J, ρ_B, and ρ_K, which are positive constants, denote the adaptation gain to be designed. Finally, the total control input u is as follows:

$$u(t) = T_m(t) = u_{eq}(t) + u_N(t) \tag{14}$$

with u_{eq} defined in (8) and u_N in (9), respectively. Then, consider the candidate Lyapunov function $V(t)$ to prove that the tracking error (5) of the system can asymptotically converge to zero for a given reference steering wheel angle with total control input u in (14). The Lyapunov function is as follows:

$$V(t) = \frac{1}{2} s(t)^2 + \lambda_2 \left[\frac{\rho_J}{2} (\Delta J_{eq})^2 + \frac{\rho_B}{2} (\Delta B_{eq})^2 + \frac{\rho_K}{2} (\Delta K_{eff})^2 \right] (J_{eq} = \hat{J}_{eq} + \Delta J_{eq},\ B_{eq} = \hat{B}_{eq} + \Delta B_{eq},\ K_{eff} = \hat{K}_{eff} + \Delta K_{eff}) \tag{15}$$

The derivative of V along the system trajectory is:

$$\dot{V}(t) = s \cdot \dot{s} + \lambda_2 [\rho_J \Delta J_{eq} (\dot{J}_{eq} - \dot{\hat{J}}_{eq}) + \rho_B \Delta B_{eq} (\dot{B}_{eq} - \dot{\hat{B}}_{eq}) + \rho_K \Delta K_{eff} (\dot{K}_{eff} - \dot{\hat{K}}_{eff})] \tag{16}$$

We assume J_{eq}, B_{eq}, and K_{eff} are constant. Therefore, we have:

$$\begin{aligned} \dot{V}(t) &= s \cdot \dot{s} + \lambda_2 [\rho_J \Delta J_{eq} (-\dot{\hat{J}}_{eq}) + \rho_B \Delta B_{eq} (-\dot{\hat{B}}_{eq}) + \rho_K \Delta K_{eff} (-\dot{\hat{K}}_{eff})] \\ &= s \cdot \dot{s} - \lambda_2 [\Delta J_{eq} s \cdot \ddot{\theta}_{sw} + \Delta B_{eq} s \cdot \dot{\theta}_{sw} + \Delta K_{eff} s \cdot \theta_{sw}] \\ &= s [\lambda_1 \dot{e} + \lambda_2 (J_{eq} \ddot{\theta}_{sw} + B_{eq} \dot{\theta}_{sw} - T_m + K_{eff} \theta_{sw} + R_{rp} F_r - T_{sw.des})] \\ &\quad - \lambda_2 [\Delta J_{eq} s \cdot \ddot{\theta}_{sw} + \Delta B_{eq} s \cdot \dot{\theta}_{sw} + \Delta K_{eff} s \cdot \theta_{sw}] \end{aligned} \tag{17}$$

Substituting the control input (14) into (18) yields:

$$\begin{aligned} \dot{V}(s(t)) &= s \{\lambda_1 \dot{e} + \lambda_2 [J_{eq} \ddot{\theta}_{sw} + B_{eq} \dot{\theta}_{sw} - T_{sw.des} + K_{eff} \theta_{sw} + R_{rp} \hat{F}_r \\ &\quad - (u_{eq} + u_N)]\} - \lambda_2 [\Delta J_{eq} s \cdot \ddot{\theta}_{sw} + \Delta B_{eq} s \cdot \dot{\theta}_{sw} + \Delta K_{eff} s \cdot \theta_{sw}] \\ &= \lambda_2 [\Delta_{un} - k \cdot s(t)\ \text{sat}(s(t)/\Phi)] \end{aligned} \tag{18}$$

Since $|\Delta_{un}| < |L_m| \le k$, $\dot{V}$ is strictly negatively defined. Thus, the system with the control input in (14) is asymptotically stable.

The aligning moment to calculate the nominal rack force, $\hat{F}_r$, constitutes most of the disturbance in steering systems. There are various ways to calculate the moment. Pacejaka [20] proposed aligning torque equations using the magic formula tire model. However, this requires many states of the tire model. On the other hand, Yih et al. [17] proposed a self-aligning torque based on a map with lateral force by multiplying the pneumatic trail as a tire slip angle. Pfeffer et al. [21] used a simple equation in which the aligning torque from the lateral tire force is calculated by multiplying the lateral tire force with the mechanical and pneumatic trail. Therefore, the nominal rack force, $\hat{F}_r$, from the aligning torque is just computed using the linear tire model in which the slip angle is assumed to exist in linear areas, and the error between the actual rack force and the model is compensated by implementing the adaption techniques as mentioned above. The nominal rack force is calculated as follows:

$$\hat{F}_r = \frac{2}{L_{arm}} \cdot \left[C_f \left(\delta_{tire} - \frac{V_y + l_f \cdot \gamma}{V_x} \right) \right] (t_m + t_p) \tag{19}$$

where L_{arm} is the steering arm length; C_f is the front lateral tire stiffness; δ_{tire} is the front tire angle; V_x and V_y are the longitudinal and lateral velocity of the vehicle, respectively; l_f is the distance between C.G. and the front tire; γ is the yaw rate of vehicle; t_m is the mechanical trail; and t_p is the pneumatic trail. Assuming that the left front tire force and the right front tire force are identical for convenience, t_m and t_p are positive constants. All vehicle state information, such as the lateral velocity or side slip angle, are very difficult or expensive to measure directly. During the last few years, significant research on estimation of vehicle side slip angle has been conducted [22–24]. A side slip angle estimator using global positioning system (GPS) measurements has been proposed [22]. Other methods for estimating the side slip angle based on a kinematic model using the extended Kalman filter [23] or a combination of kinematic and dynamics model using lateral acceleration and yaw rate motion measurements [24] have been evaluated. It was assumed that the side slip angle of a vehicle can be obtained by a side slip angle state estimator that was designed in previous research [22–24]. Although this simple rack force from the linear tire model using estimated state or nominal parameter is not accurate compared to actual rack force, the error is compensated through adapting the effective compliance parameter. Therefore, the effective compliance force, $\hat{K}_{eff}\theta_{sw}$, with the nominal rack force, $\hat{F}_r$, approaches the actual rack force. For this paper, the vehicle parameters are shown in Table 1.

3.3. Steering states estimation with a Kalman filter

It is essential to filter signals to improve controller performance, and a Kalman filter and a low-pass filter are utilized in this research. The steering system is equipped with a steering angle, as well as steering wheel torque sensors. Ideally, this signal should be reasonably clean since high-frequency noise tends to worsen with differentiation, and heavy filtering might induce a lag in the signal [17]. Therefore, the solution to obtain $\dot{\theta}_{sw}$, $\ddot{\theta}_{sw}$, and T_{sw} is to pass the measured steering wheel angle $\theta_{sw,\ sensor}$ with the Kalman filter and $T_{sw,\ sensor}$ through the first-order filter. The Kalman filter state vector, $\mathbf{x}$, and the state space equation are defined as follows (20), (21):

$$\mathbf{x} = [\theta_{sw} \quad \dot{\theta}_{sw} \quad \ddot{\theta}_{sw}]^T \tag{20}$$

$$\mathbf{x}[k+1] = \mathbf{A}_d \mathbf{x}[k] + \mathbf{w}[k] \tag{21}$$

where,

$$\mathbf{A}_d = \begin{bmatrix} 1 & \Delta T & \Delta T^2/2 \\ 0 & 1 & \Delta T \\ 0 & 0 & 1 \end{bmatrix}$$

where $\mathbf{w}[k]$ denotes the process noise that is assumed to have been drawn from a zero-mean multivariate normal distribution with covariance $\mathbf{Q}_k$; and ΔT is the sampling time (0.001 s in this study). The state transition model is designed using the kinematics relationship of

Table 1
Vehicle parameters.

Parameter	Value	Units
l_f	107.95	mm
C_f	24.5	kN
t_m	2.6	cm
t_p	1.4	cm

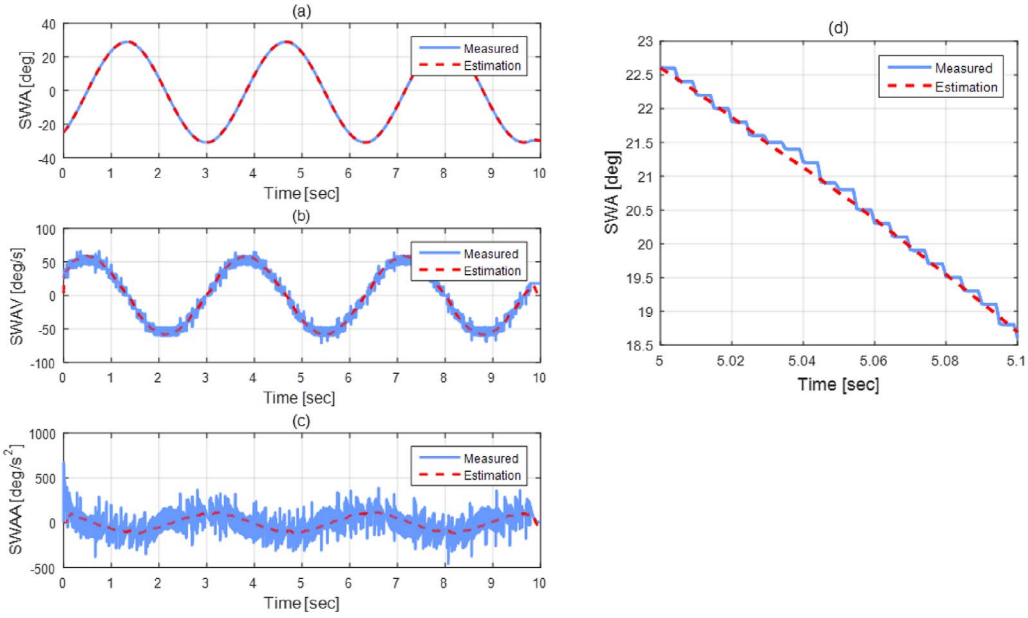


Fig. 5. Steering states estimation results with hard-ware-in-loop simulation. (a) Steering wheel angle. (b) Steering angular velocity. (c) Steering angular acceleration. (d) Discrete steering wheel angle.

steering motion. Therefore, since the position-velocity and the velocity-acceleration relationship is correctly a differential relationship, all of the matrix elements are zero, except for the third row and third column value of the covariance $\mathbf{Q}_k$. For this reason, $\mathbf{Q}_k^{3,3} = 100$ is chosen for implementation.

Only the steering angle is assumed to be in a measurable state, and thus the measurement model can be defined as follows:

$$\mathbf{z}[k] = \mathbf{H}\mathbf{x}[k] + \mathbf{v}[k] \tag{22}$$

where,

$$\mathbf{H} = [1 \quad 0 \quad 0]$$

where $\mathbf{v}[k]$ is the observation noise that is assumed to be the zero mean Gaussian white noise with covariance $\mathbf{R}_k$. As can be seen from the estimation results below in Fig. 5(a), the angle sensor used in the steering system does not possess much noise. Therefore, $\mathbf{R}_k^{1,1} = 0.1$ is chosen for implementation.

The above process model and measurement model are used with the Kalman filter to estimate steering angular velocity and steering angle acceleration. The sequence for the Kalman filter, which recursively estimates the states at each time-step, is composed of time and measurement updates (23), (24):

- *Time update*

$$\begin{aligned} \bar{\mathbf{x}}[k] &= \mathbf{A}_d[k]\cdot\hat{\mathbf{x}}[k-1] \\ \bar{\mathbf{P}}[k] &= \mathbf{A}_d[k]\cdot\hat{\mathbf{P}}[k-1]\cdot\mathbf{A}_d{}^T[k] + \mathbf{Q}_k \end{aligned} \tag{23}$$

- *Measurement update*

$$\begin{aligned} \mathbf{K}[k] &= \bar{\mathbf{P}}[k]\cdot\mathbf{H}^T\cdot(\mathbf{H}\cdot\bar{\mathbf{P}}[k]\cdot\mathbf{H}^T + \mathbf{R}_k)^{-1} \\ \hat{\mathbf{P}}[k] &= (\mathbf{I} - \mathbf{K}[k]\cdot\mathbf{H})\cdot\bar{\mathbf{P}}[k] \\ \hat{\mathbf{x}}[k] &= \bar{\mathbf{x}}[k] + \mathbf{K}[k]\cdot(\mathbf{z}[k] - \mathbf{H}\cdot\bar{\mathbf{x}}[k]) \end{aligned} \tag{24}$$

where $\mathbf{P}[k]$, $\mathbf{K}[k]$, and $\hat{\mathbf{x}}[k]$ are the error covariance, Kalman gain, and estimated state, respectively, at the k-th time-step. The steering states are estimated well, as shown in Fig. 5. Fig. 5(a) and (d) show the results for the steering wheel angle. As shown in Fig. 5(d), the measured steering wheel angle represents the discretized value from the sensor of the EPAS system. Fig. 5(b) and (c) represent the steering angular velocity and the angular acceleration from the sensor and the estimation, respectively. The sensor uses a derivative of the measured steering wheel angle and low-pass filter to reduce high-frequency values. Since the discretized values are directly differentiated, the high-frequency components remain as shown in Fig. (b) and (c). If a low-pass filter is used to reduce the high-frequency values, the desired performance will not be expected due to the phase delay problem. On the other hand, the estimated value indicates that the high-frequency component has been removed.

4. Simulation results

Computer simulations were conducted with Matlab/Simulink and Carsim co-simulation to evaluate the proposed EPAS torque control algorithm using adaptive sliding mode control. Weave and transition open-loop tests were performed to investigate the performance of the proposed algorithm. The weave and transition tests are simulated on a dry road (friction coefficient, 0.85). Open-loop command determines the steering wheel angle. In the case of weave, the steering wheel angle input is 30°, 0.3 Hz. The vehicle speed is set to be constant at 60 km/h The driving situation at 60 km/h shows typical steering characteristics well because the case has the moderate characteristic of rack force (hysteresis and rate of rack force) according to Fig. 2. In the case of transition, the steering wheel angular velocity input is 5°/s. The vehicle speed is also set to be constant at 60 km/h. All of the nominal parameter values and control gain used for the computer simulation study are described in Table A1 in the Appendix.

Fig. 6 shows the weave simulation results. Fig. 6(a) presents the steering wheel angle input. Fig. 6(b) shows the control input current from the controller. Fig. 6(c) presents the steering wheel torque with respect to steering wheel angle. The reference is the target steering wheel torque from the target steering wheel torque map, and the actual is the steering wheel torque of a driver. It was observed that the actual tracked the target well. Fig. 6(f) also shows the tracking error plot between the steering wheel torque and the target steering wheel torque. Time histories of the three adaptation parameters, the moment of inertia, $\hat{J}_{eq}$, the damping coefficient, $\hat{B}_{eq}$, and the effective compliance, $\hat{K}_{eff}$, are shown in Fig. 6(d). The moment of inertia and damping coefficient reveal that there is not much variation in time because the values are known in advance as system parameters. However, effective compliance varied according to the situation. Fig. 6(e) presents the actual rack force compared with nominal rack force, $\hat{F}_r$, and adapted

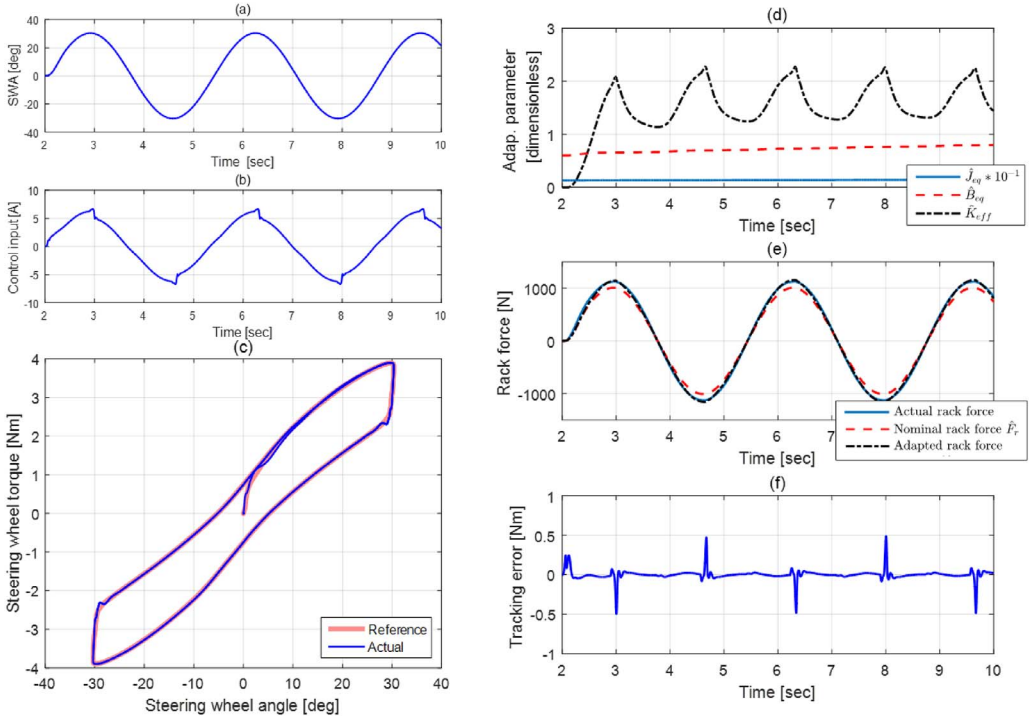


Fig. 6. 30° weave test at 60 km/h. (a) Steering wheel angle. (b) Control input current from controller. (c) Steering wheel torque vs. steering wheel angle. (d) Adaptation parameters. (e) Comparison of rack force. (f) Tracking error between steering wheel torque and target steering wheel torque.

rack force. The adapted rack force is the sum of the nominal rack force and the effective compliance force, $\widehat{K}_{eff}\theta_{sw}/R_{rp}$. It can be seen that the adapted rack force tracks the actual rack force well.

Fig. 7 presents the transition test results. As the above weave test, Fig. 7(a)–(f) show steering wheel angle, control input, steering wheel torque with respect to steering wheel angle, adaptation parameters, rack forces, and tracking error. As shown in Fig. 7(c), it was also observed that the actual tracked the target well. The actual rack force, nominal rack force, $\widehat{F}_r$, and adapted rack force are shown in Fig. 7(e).

The adapted rack force is the sum of the nominal rack force and the effective compliance force, $\widehat{K}_{eff}\theta_{sw}/R_{rp}$. It can be seen that the adapted rack force tracks the actual rack force well, as the above weave simulation.

A comparison of the proposed ASMC with PID 1 (P = 2, I = 200, D = 0.03) and PID 2 (P = 2, I = 100, D = 0.03) control is conducted on the weave test to investigate the steering wheel torque tracking performance. The simulation is conducted with a 60 kph, 30° steering wheel angle input with 0.3 Hz. As can be seen in Fig. 8(e), the ASMC

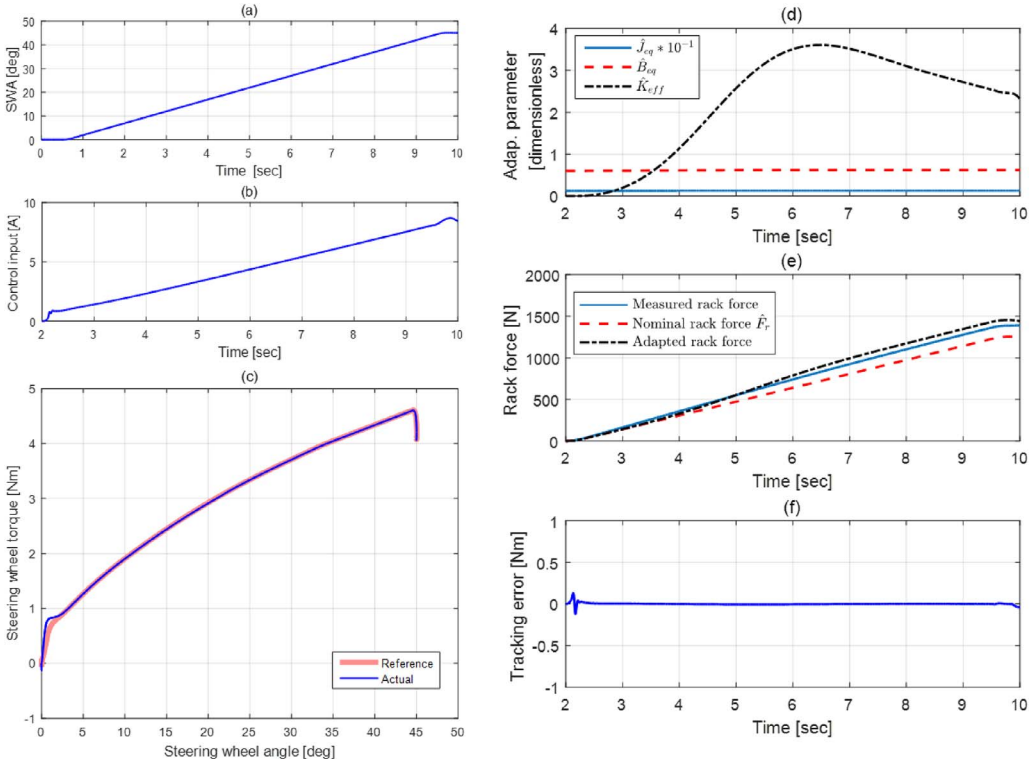


Fig. 7. 5°/s transition test at 60 km/h. (a) Steering wheel angle. (b) Control input current from controller. (c) Steering wheel torque vs. steering wheel angle. (d) Adaptation parameters. (e) Comparison of rack force. (f) Tracking error between steering wheel torque and target steering wheel torque.

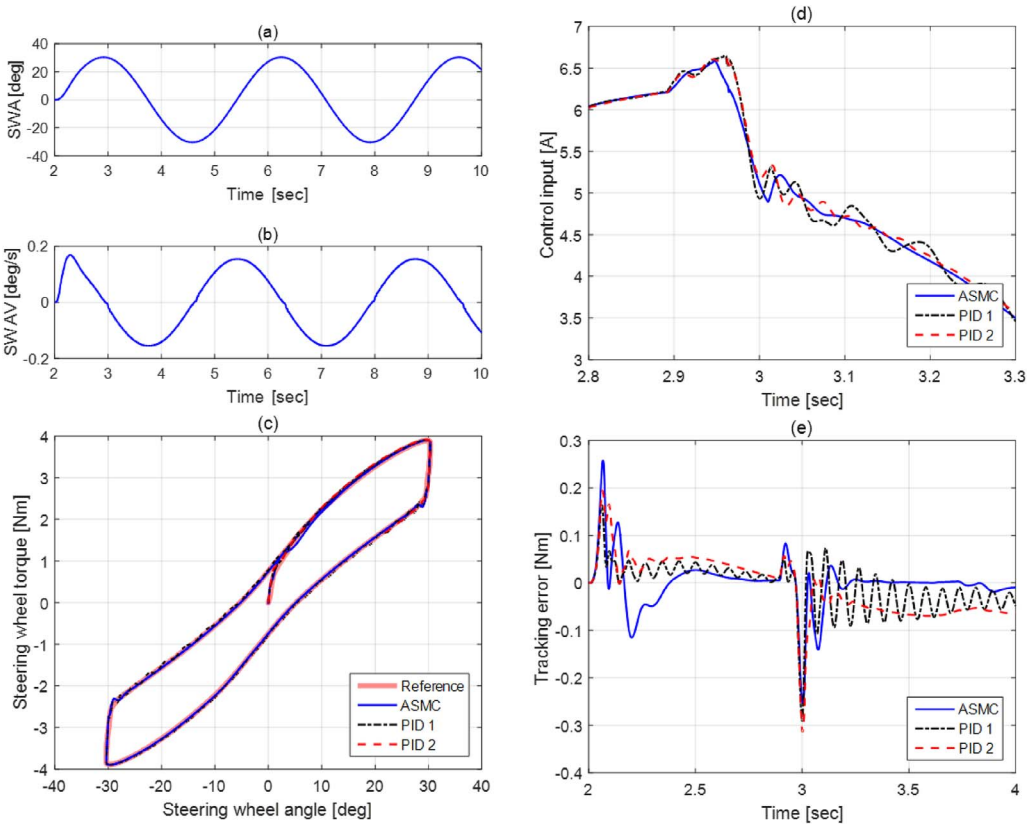


Fig. 8. Comparison of the proposed ASMC with PID control. (a) Steering wheel angle. (b) Steering wheel angular velocity. (c) Steering wheel torque vs steering wheel angle. (d) Control input current (e) Tracking error between steering wheel torque and target steering wheel torque.

has the smallest steady-state error of these. A PID 1 simulation is conducted to reduce the steady-state error using a larger I gain than that for PID 2. Although the steady-state error for PID 1 has been reduced, this case has a more massive oscillation than PID 2. The reason for this is that the steering wheel torque tracking error increased abruptly during the return, as shown in Fig. 8(e). Therefore, an oscillation occurs if a larger gain was adopted. In contrast, the proposed ASMC tracks the target steering wheel torque well in all situations, as seen in Fig. 8(e) for the ASMC.

The nominal rack force, $\hat{F}_r$, varies from 50% to 150% to analyze the robustness of the performance with respect to the nominal rack force parameter. The calculated value from Eq. (19) depicts the reason behind the changing toe nominal rack. However, this may be inaccurate due to parameter uncertainties, such as pneumatic trail, mechanical trail, steering arm length, and lateral tire stiffness uncertainty. All other conditions are identical to those in the above simulation. Fig. 9 shows the root mean square (RMS) tracking error between the steering wheel torque and the target steering wheel torque with respect to $\hat{F}_r$ for the presence and absence of the adaption. The variation in performance for the presence of the adaptation is much smaller than that with the absence. These results imply that the proposed adaptation sliding model control is very useful in tracking steering wheel torque against parameter uncertainty.

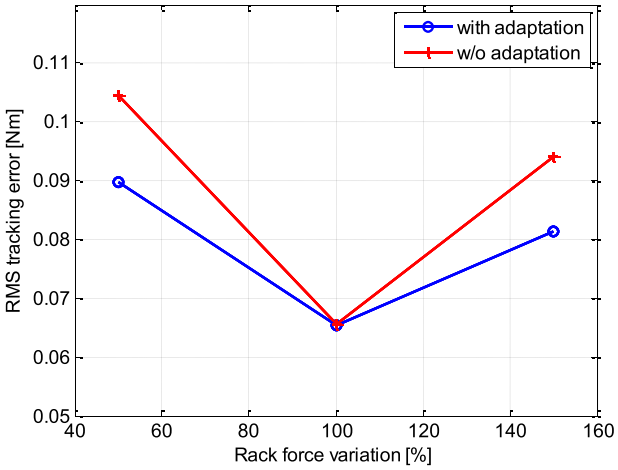


Fig. 9. Comparison of the RMS tracking error with and without adaptation.

5. Hardware-in-the-loop test results

The proposed control algorithm was investigated by the hardware-in-the-loop-simulation (HILS). As shown in Fig. 10, the HILS consists of two parts: a software part and a hardware part. In the software part, Carmaker from IPG Automotive was used to implement the full vehicle dynamics model using the real-time simulator (DS1006) from dSPACE and micro-autobox (1401/1511/1513) to implement the proposed EPAS control algorithm from dSPACE. The hardware part comprises five components. The steering robot generates the accurate steering input. The hydraulic cylinder unit (maximum output 15-kilonewton) from MTS Systems Corporation creates the rack force from the vehicle model. EPAS ECU controls the motor using the current input from the proposed algorithm. The rack displacement sensor is transferred to the dynamic model by measuring the displacement rack. The TAS sensor measures the torque and angle, and transmits the information to the autobox. The parameter values and control gains are listed in Table A1 in the Appendix.

As previously conducted with the computer simulation, the weave and transition tests are simulated on a dry road (friction coefficient, 0.85). Open-loop command determines the steering wheel angle, as the above computer simulation tests. In the case of weave, the steering wheel angle input is 30° and 0.3 Hz. The vehicle speed is set to be constant at 60 km/h. In the case of transition, the steering wheel angular velocity input is 5°/s. The vehicle speed is also set to be constant

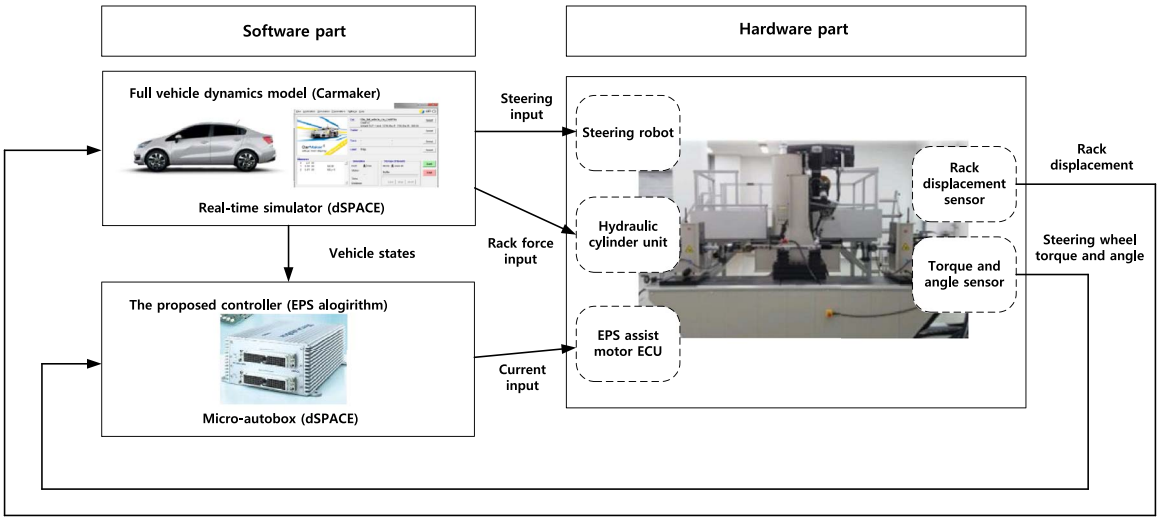


Fig. 10. Schematic diagram of the hardware-in-the-loop-simulation (HILS).

at 60 km/h.

Fig. 11 shows the HILS results of the weave test. The results of the HILS simulation are arranged in parallel with the results in Fig. 6 of the computer simulation. It was also observed that the actual tracked the reference well, as shown in Fig. (c) and (f). Unlike the computer simulation, in the weave test, effective compliance was slightly smoother, as shown in Fig. 11(d). The adapted rack force tracks the actual rack force well, as with the computer simulation in Fig. (e). As the effective compliance changed, it can be seen that the error gradually decreased, as shown in Fig. 11(f).

Fig. 12 shows the HILS transition test results. The results of the HILS simulation are arranged in parallel with the results in Fig. 7 of the computer simulation. It was also observed that the actual tracked the reference well, as shown in Fig. (c) and (f). In addition, it can be seen that the adapted rack force tracked the actual rack force well, as shown in Fig. (e). Although the equivalent moment of inertia, $\hat{J}_{eq}$, the equivalent damping, $\hat{B}_{eq}$, and the effective compliance, $\hat{K}_{eff}$, are chosen as adaptation parameters to guarantee robustness to parameter uncertainty, it was confirmed that the effective compliance was dominant compared to others, as can be seen in the above results.

According to Jang et al. [5], the objective parameters used in the steering performance evaluation are presented. In the weave test results, the steering angle vs. steering wheel torque graph can be utilized to derive such parameters as on-center stiffness, steering friction and angle hysteresis, as can be seen in Table 2. Here, the on-center stiffness is the gradient at zero angles, the steering friction is ordinate deadband, and the angle hysteresis is abscissa deadband. In addition, average steering friction and off-center stiffness were used in the transition test results, as can be seen in Table 3. The average steering friction is average torque 0° to 1°, and the off-center stiffness is an average gradient of 5° to 15°. Computer simulation and HILS target results are different. This is because the target determined by the steering wheel

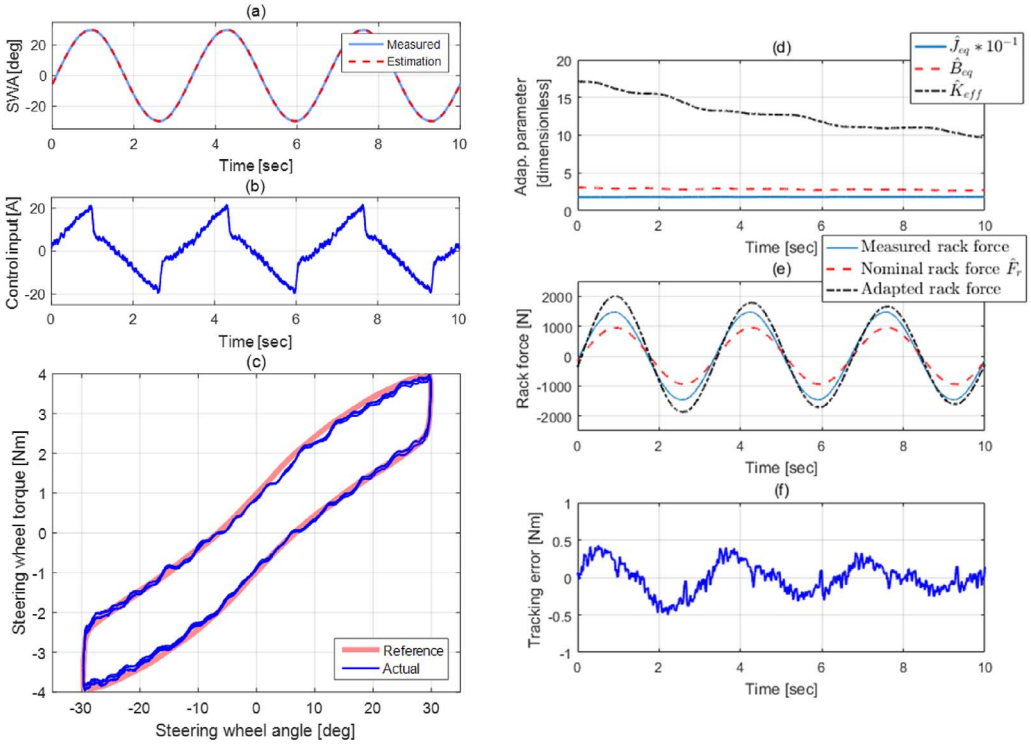


Fig. 11. 30° weave test at 60 km/h. (a) Steering wheel angle. (b) Control input current from controller. (c) Steering wheel torque vs. steering wheel angle. (d) Adaptation parameters. (e) Comparison of rack force. (f) Tracking error between steering wheel torque and target steering wheel torque.

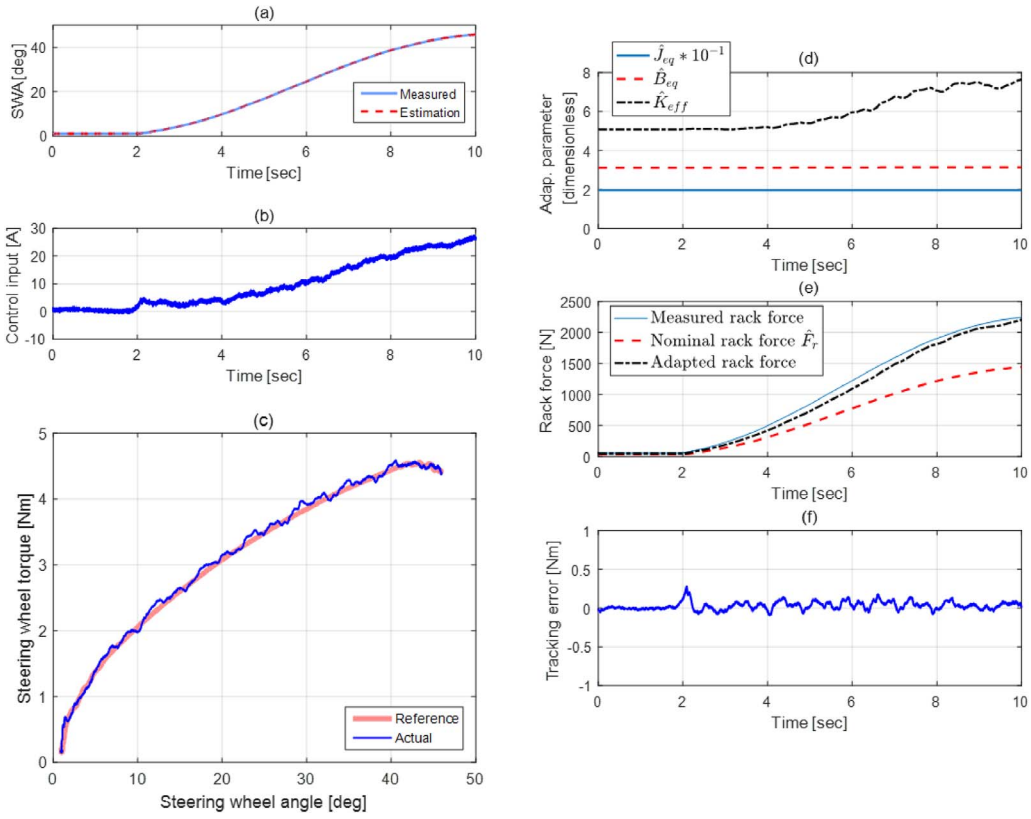


Fig. 12. 5°/s transition test at 60 km/h. (a) Steering wheel angle. (b) Control input current from controller. (c) Steering wheel torque vs. steering wheel angle. (d) Adaptation parameters. (e) Comparison of rack force. (f) Tracking error between the steering wheel torque and the target steering wheel torque.

Table 2
Comparison of computer simulation and HILS results using objective parameters with the weave test.

	Unit	Computer simulation		HILS	
		Target	Actual	Target	Actual
On-center stiffness	Nm/deg	0.1577	0.1693	0.1591	0.1613
Steering friction	Nm	1.8563	1.8869	2.2316	1.8962
Angle hysteresis	deg	10.515	11.006	13.1191	12.664

Table 3
Comparison of computer simulation and HILS results using objective parameters with the transition test.

	Unit	Computer simulation		HILS	
		Target	Actual	Target	Actual
Average steering friction	Nm	0.4207	0.4172	0.5296	0.5410
Off-center stiffness	Nm/deg	0.1141	0.1142	0.1206	0.1225

angle and the angular velocity changes when the simulation environment is changed. In the weave test, the average error of the parameters of the computer simulation is approximately 4.5%, and the average error of the HILS is approximately 6.6%. In the transition test, the mean error of the parameters of the computer simulation is approximately 0.5%, and the mean error of the HILS is approximately 1.9%. Based on the average errors, the experimental results are satisfactory.

Fig. 13 presents the comparison of HILS weave test results by vehicle speed. Fig. 13(a) shows the steering wheel angle vs. steering wheel torque graph. In order to reflect the rack force characteristics (Fig. 2) in the actual vehicle, the target steering wheel torque map was constructed so that the steering stiffness increases as the vehicle speed increases (40 kph: 0.1030 N m/°, 60 kph: 0.1591 N m/°, 80 kph: 0.1845 N m/°). On the contrary, the magnitude of the steering friction decreases with speed (40 kph: 3.4306 N m, 60 kph: 2.2316 N m, 80 kph: 1.427 N m). Fig. 13(b) shows the error values of the target for each speed and the steering wheel torque. It can be seen that the tracking performance is excellent. Fig. 13(c) presents the adapted rack force, which is the sum of the nominal rack force and the effective compliance force. Fig. 13(d) shows the error value between the actual rack force and the adapted rack force. The estimated performance is satisfactory.

Fig. 14 shows a comparison of HILS transition test results by steering angular velocity. Fig. 14(a) presents the steering wheel angle. Fig. 14(b) shows steering wheel angle vs. steering wheel torque. The target steering wheel torque map was designed so that the average steering torque increases as the steering angular velocity increases (5°/s: 0.5296 N m, 45°/s: 0.7784 N m, 90°/s: 1.1082 N m), and also increase the off-center stiffness (5°/s: 0.1206 N m/°, 45°/s: 0.1728 N m/°, 80 kph: 0.2045°/s). Fig. 14(c) presents the error values of the target for each speed and the steering wheel torque. It can be seen that the tracking performance is good. Fig. 14(d) shows the adapted rack force, which is the sum of the nominal rack force and the effective compliance force. Fig. 14(e) presents the error value between actual rack force and adapted rack force. The estimated performance is satisfactory.

5. Conclusion

A target steering wheel torque tracking algorithm has been presented for application to an electric-power-assisted-steering (EPAS) system. The tracking algorithm has been developed to overcome a major challenge of the conventional method, which is time-consuming tuning to achieve the desired steering feel. A target steering wheel torque map defined from the steering wheel angle, steering angular velocity, and vehicle speed gives drivers the desired steering feel. The target steering wheel torque was tracked with an adaptive sliding mode control (ASMC) using an integral augmented sliding surface. The equivalent moment of inertia damping coefficient and effective compliance were adopted to achieve robust tracking performance. Weave open-loop tests were conducted to identify the characteristic of rack

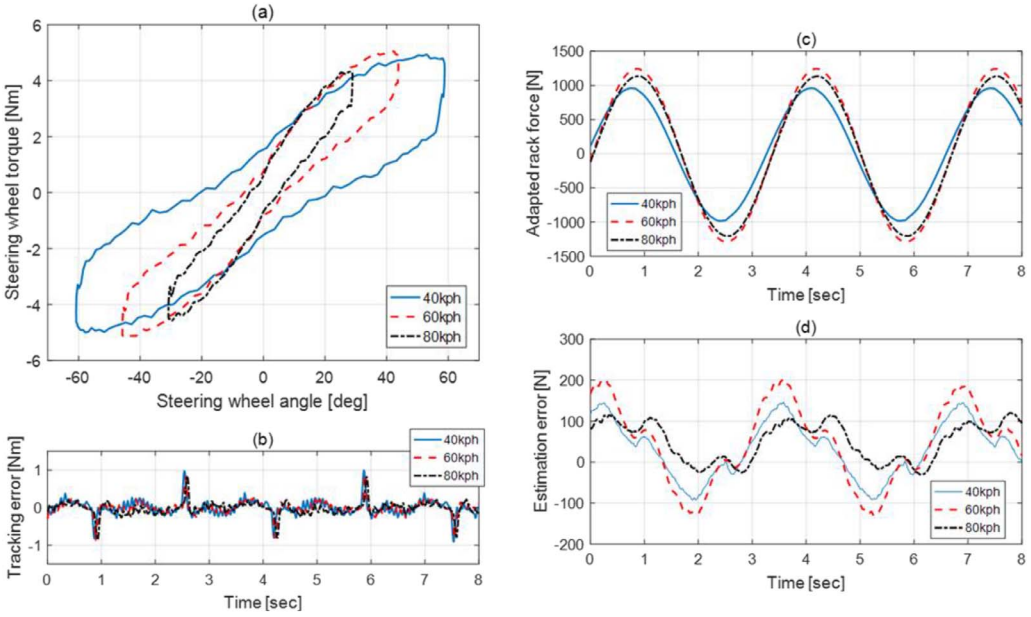


Fig. 13. Comparison of HILS weave test results by different vehicle speeds. (a) Steering wheel angle vs. steering wheel torque. (b) Tracking errors between target and steering wheel torque. (c) Adapted rack force. (d) Estimation errors between actual rack force and adapted rack force.

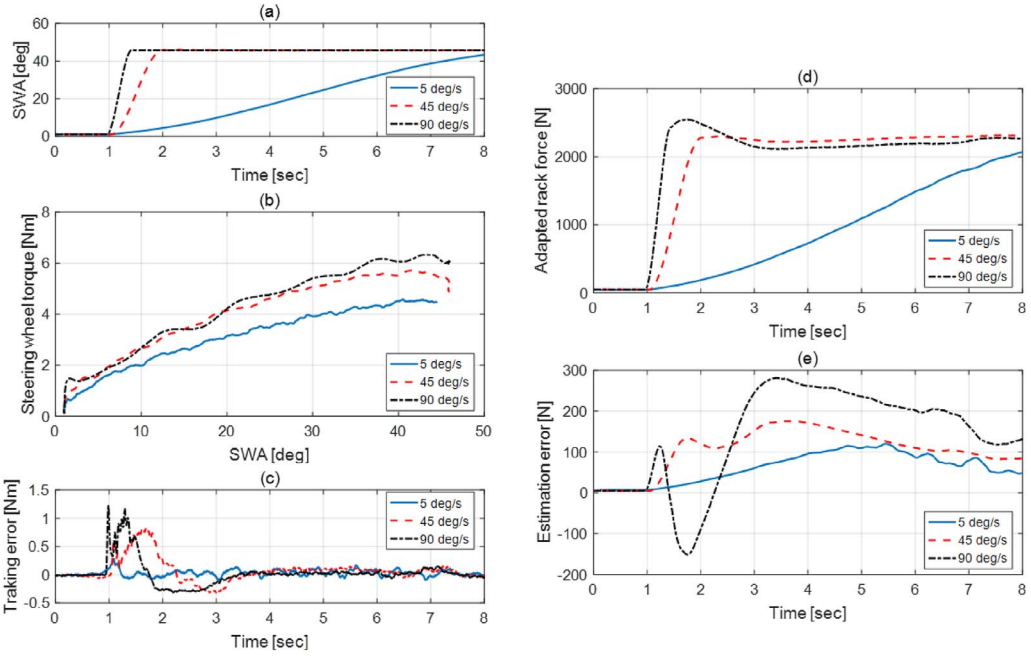


Fig. 14. Comparison of HILS transition test results by different steering angular velocities. (a) Steering wheel angle. (b) Steering wheel angle vs. steering wheel torque. (c) Tracking errors between target and steering wheel torque. (d) Adapted rack force. (e) Estimation errors between actual rack force and adapted rack force.

force. As a result, rack force has linearity in a stable range. Therefore, the rack force can be assumed to be linear using effective compliance and was selected as an adaption parameter.

Simulation results and hardware-in-the-loop system (HILS) tests were performed to evaluate the proposed ASMC to track the target steering wheel torque. The simulations are performed with weave and transition tests. Effective compliance plays a role in compensating for the error between the nominal rack force and the actual rack force, as can be seen in the simulation results. In addition, it was confirmed that the effective compliance was dominant compared to other factors, as can be seen above from the results of experiments. The proposed EPAS control algorithm achieved satisfactory steering wheel torque tracking performance.

Vehicle tests and evaluation of the proposed steering wheel torque tracking algorithm constitute the topics of our future research.

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Appendix A

Table A1
Control design parameters: computer simulation and HILS.

Parameter	Value		Units	Description
	Computer simulation	HILS		
$\hat{J}_{eq}$	0.0129	0.2	$kg \cdot m^2$	Nominal value of moment of inertia
$\hat{B}_{eq}$	0.6	3.1	Nm/(rad/s)	Nominal value of damping coefficient
$\hat{K}_{eff}$	0	5	Nm/rad	Nominal value of effective compliance
R_{rp}	0.012	0.012	m	Gear ratio between rack and column
λ_1	0.1	1	–	Error gain
λ_2	7	1	–	Integral error gain
k	4	4	–	Switching control gain
Φ	0.5	0.5	–	Boundary layer of switching control
ρ_J	100	100	–	Adaptation gain for inertia
ρ_B	1	1	–	Adaptation gain for damping
ρ_K	0.01	0.5	–	Adaptation gain for effective compliance

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